

Smart Business Analytics Management System

Problem Statement

Develop a Smart Business Analytics Management System using Python. The system must integrate Object-Oriented Programming, File Handling (Text & CSV), NumPy for numerical analysis, Pandas for structured reporting, and Matplotlib for visualization. All components must work together in a single application.

PART A: Object-Oriented Design

- Create a class 'Employee' with Employee ID, Name, and Monthly Sales (6 months).
- Include methods to calculate Total Sales, Average Sales, and determine Performance Grade.
- Implement inheritance using 'SeniorEmployee' and 'JuniorEmployee' classes.
- Apply polymorphism for Bonus Calculation method.

PART B: File Handling

- Store employee data in employees.txt and employees.csv files.
- Read data from CSV file.
- Implement proper exception handling for File Not Found, invalid numeric values, and incorrect format.

PART C: NumPy Analysis

- Convert sales data into NumPy arrays.
- Calculate Overall Mean Sales, Standard Deviation, and Highest Sales Month.
- Identify employees below company average.

PART D: Pandas Reporting

- Load CSV data into a DataFrame.
- Add calculated columns: Total Sales, Average Sales, and Grade.
- Filter Top 3 Employees and Low Performers.
- Display summary statistics.

PART E: Data Visualization

- Generate a Bar Chart for Employee vs Total Sales.
- Generate a Line Graph for Monthly Sales Trend.
- Generate a Pie Chart for Performance Distribution.